

Sara Abid

+92 322 403 8758 · saraabidhussain12@gmail.com · github.com/SaraAbidHussain · Lahore, Pakistan

EDUCATION

Bachelor of Science in Computer Science

Aug 2024 – May 2028

Information Technology University of the Punjab (ITU), Lahore, Pakistan

CGPA: **3.65 / 4.00** (90.54%) · 51 Credit Hours Completed · Relevant: AI, Data Structures, Algorithms, Database Systems, OOP, Linear Algebra, Probability

PROJECTS

IntelliCare-ICU — Intelligent ICU Patient Monitoring System

C++17 · AWS EC2 · REST API · github.com/SaraAbidHussain/Intellicare-ICU-System

- Designed 7 custom data structures from scratch: B-Tree, Hash Table, Min-Heap, KD-Tree (5D clustering), Drug-Interaction Graph (DFS safety checks), Circular Buffer, Sliding Window
- Built 15+ REST API endpoints; deployed live on AWS EC2 with sub-2ms query times and under 50ms API response on t2.micro

AI Research Paper Writing Assistant

Python · Hugging Face Transformers · PyTorch · Flask

- Building an end-to-end NLP pipeline integrating LLMs to help researchers draft and structure academic papers; leading full SDLC from SRS documentation to model integration

Sign Language to Urdu Translator

Python · Computer Vision · NLP · Real-Time Gesture Recognition

- Building a real-time translator combining hand gesture recognition with NLG to enable communication between hearing-impaired and hearing individuals

Smart Gym and Fitness Center Management System

Flutter Web · Flask · PostgreSQL 15 · Docker Compose · AWS EC2 · JWT

- Built 4 role-based dashboards; designed 15-table PostgreSQL schema with triggers, views, and indexes reducing query times by 15x; deployed on AWS EC2

ML from Scratch — Linear and Logistic Regression

Python · NumPy · Scikit-learn · Matplotlib

- Implemented gradient descent with L2 norm convergence; logistic regression with 5-fold CV and threshold tuning (0.3–0.7); benchmarked against Scikit-learn baselines

PCA-Based Face and Expression Recognition

Python · NumPy · Machine Learning

- Built full pipeline: covariance matrix computation, eigenspace projection, expression classification; extended to multi-class facial expression recognition

PostgreSQL Extension — Custom Data Types and Operators

C · PostgreSQL Internals

- Developing a PostgreSQL extension in C introducing custom data types and operators, working at database internals level

Chess Engine and Metal Slug Replica

C++ · Raylib · github.com/SaraAbidHussain/Chess

- Chess: complete legal move enforcement, check/checkmate detection, state persistence, Raylib GUI — Metal Slug: sprite animation, collision detection, enemy AI, game-loop architecture

TECHNICAL SKILLS

Languages:	C++, Python, C, Dart, SQL, JavaScript/HTML/CSS, C#
AI / ML:	PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, Gradient Descent, PCA, Classification, Object Detection
Web / Mobile:	Flutter, Flask, REST APIs, JWT Authentication, Socket Programming
DevOps:	AWS EC2, Docker, Docker Compose, Nginx, Linux, Git, GitHub
Databases:	PostgreSQL, MySQL, Oracle — schema design, ACID transactions, triggers, views, indexing

EXPERIENCE

Teaching Assistant — Modern Mathematics for CS

Seed Programming, Lahore

- Taught discrete mathematics, combinatorics, logic, graph theory — core CS/AI foundations

Event Management Co-Lead — GDG ITU

Information Technology University, Lahore

- Co-leads developer workshops, speaker sessions, and community outreach for GDG ITU chapter

Summer 2025